(ii) Since the point D lie vertically above $C(0,2)$. The x coordinate of D is 0.
Since any point on North Street has x coordinate value 0.
The equation of North Street is $x = 0$

(iii) To find equation of Cross Road.

Center of circular garden M is at $(7, 7)$, A is $(3, 10)$

We first find slope of MA, which we call m_1

Thus $m_1 = \frac{10-7}{3-7} = \frac{-3}{4}$.

Since the Cross Road is perpendicular to MA, if m_2 is the slope of the Cross Road then, $m_1 m_2 = -1$ gives $\frac{-3}{4} m_2 = -1$ so $m_2 = \frac{4}{3}$.

Now, the cross road has slope $\frac{4}{3}$ and it passes through the point $A(3,10)$.

The equation of the Cross Road is $y - 10 = \frac{4}{3}(x - 3)$

$$3y - 30 = 4x - 12$$

Hence, $$4x - 3y + 18 = 0$$

(b) (i) If D is $(0,k)$ then D is a point on the Cross Road.
Therefore, substituting $x = 0$, $y = k$ in the equation of Cross Road,
we get, $0 - 3k + 18 = 0$
Value of $k = 6$
Therefore, D is $(0, 6)$

(ii) To find E, let E be $(q,2)$
Put $y = 2$ in the equation of the Cross Road,
we get, $4q - 6 + 18 = 0$

$$4q = -12 \quad \text{gives} \quad q = -3$$

Therefore, The point E is $(-3,2)$

Thus the Cross Road meets the North Street at $D(0, 6)$ and East Avenue at E $(-3,2)$.

Progress Check Fill the details in respective boxes

S.No.	Equation	Slope	x intercept	y intercept
1	$3x - 4y + 2 = 0$			
2	$y = 14x$			0
3			2	–3

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